GEOSPATIAL DECISION INTELLIGENCE

Istanbul Site Selection & Market Expansion

Detailed methodology, validation, governance and operating controls for an auditable synthetic retail decision-support platform.

MarmaraMart - fictional retail chain
Version 1.0.0 - 29 July 2026

All commercial outcomes are deterministic synthetic data. No personal or sensitive data is used. This report is decision support, not investment advice.

PROJECT STATUS / READY FOR REVIEW - NOT PUBLISHED

01 / Executive summary

The platform converts Istanbul retail expansion from a radius-based map exercise into an auditable site-and-portfolio decision. It links network accessibility, microzone demand, competitive pressure, customer diversion, economics and execution risk, then separates transparent site ranking from constrained portfolio allocation.

5,965	24	81.236	4
H3 resolution-8 microzones	candidate locations	top C24 score / 100	base selected sites

TRY 93.382m	1,329,291	23.125%	TRY 28.826m
base budget used	incremental 10-minute population	total modeled market coverage	base expected portfolio EBIT

Analytical recommendation: approve field validation and commercial diligence for C24 Ikitelli, C18 Sultanbeyli, C17 Sancaktepe and C07 Eyupsultan. Do not approve capital from these outputs alone.

C24 leads the location score at 81.236/100 and has low modeled cannibalization, but its controlled three-year ROI is -16.0%. C18 is the only base-selected site with positive controlled three-year ROI (+14.7%). The gap between demand strength and site economics is an explicit decision gate.

02 / Contents and review path

Recommended review sequence

- Challenge the decision framing and synthetic-data boundary.
- Inspect network reach before accepting any population number.
- Review SHAP demand explanations separately from AHP score contributions.
- Test score weight sensitivity and scenario constraints.
- Validate individual site economics before considering portfolio synergy.
- Complete field, routing, legal and commercial diligence before capital approval.

Traceability spine: configuration -> processed data -> quality ledger -> model/OOF predictions -> SHAP -> AHP contributions -> optimizer selection -> dashboard/report.

03 / Decision architecture

Layer	Primary method	Audit evidence
Spatial foundation	WGS84, EPSG:32635, H3 resolution 8	GeoJSON, CRS and geometry checks
Demand and access	H3 network, Huff/gravity, spatial CV	catchments, OOF predictions, SHAP
Explainable score	AHP plus direction-adjusted min-max	weights and factor contributions
Portfolio allocation	binary maximum coverage	solver status, budget, distance, coverage
Delivery	API, PostGIS, maps, Excel, PBIP	contracts, views, runbook and manifest

Site rank and portfolio choice are intentionally distinct. A high-ranked candidate can be excluded because of overlap, distance conflict, capacity or budget. Conversely, a lower-ranked site can improve incremental coverage at portfolio level.

Component responsibilities

- Geo quality gates stop invalid geometry or CRS before analytical joins.
- Network catchments count unique cells reached within time thresholds.
- Demand predictions and score contributions are persisted separately.
- Optimization consumes approved scenario assumptions and exposes terminal status.
- Every user-facing artifact is generated from pipeline output rather than typed KPIs.

04 / Data provenance and synthetic design

The project uses approximate public-place coordinates only to frame a metropolitan analysis. Population, income, purchasing power, rent, traffic, store performance, sales, EBIT and investment costs are generated deterministically with seed 20260729.

Layer	Provenance	Privacy / governance
Geographic anchors	Approximate public places	No customer or household address
Commercial outcomes	Deterministic generator	Synthetic; reproducible
Network indicators	H3 geometry plus synthetic indices	No device traces
Basemap	OpenStreetMap tiles	Attribution included
Decision outputs	Local pipeline	Human review required

The absence of personal data removes person-level privacy exposure, but geographic allocation bias remains possible through income, centrality and current-estate proxies.

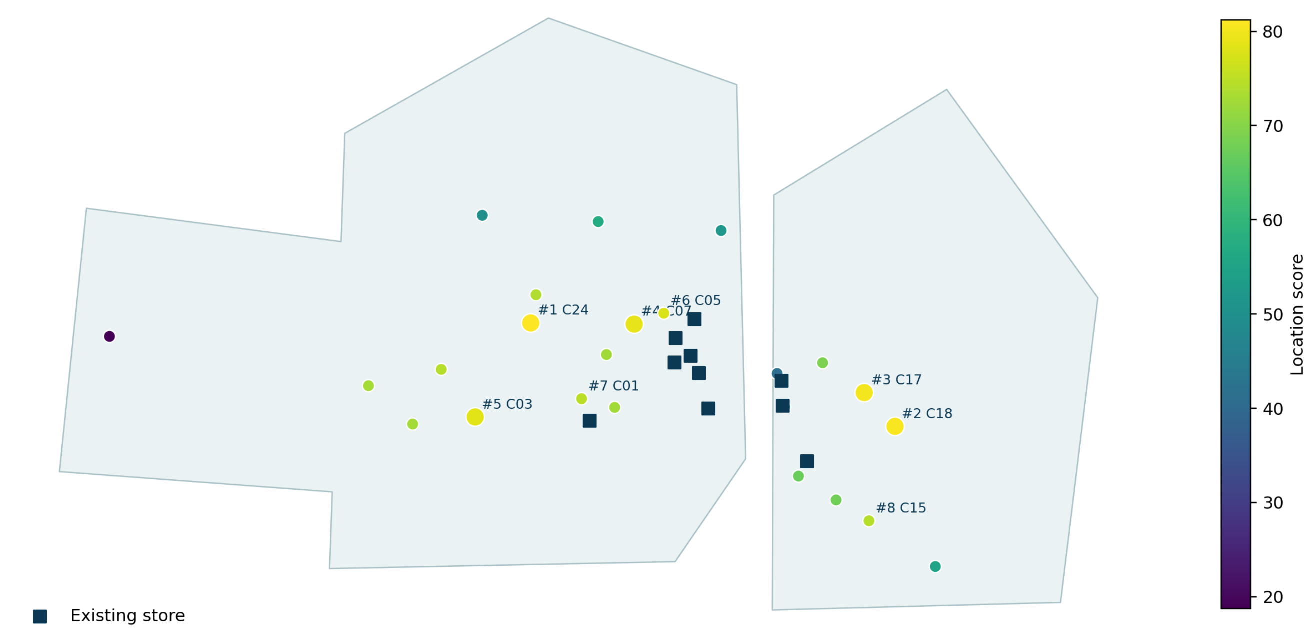
The analytical footprint is a project-specific approximation and must not be represented as an official administrative boundary. Exact parcel coordinates are a field-validation input, not a model output.

05 / CRS, geometry and H3 microzones

All exchange geometry is stored in WGS84 (EPSG:4326). Distance and area calculations use UTM zone 35N (EPSG:32635). This prevents the common error of calculating meters directly from longitude/latitude degrees.

EPSG:4326	EPSG:32635	H3 r8	5,965
storage / web exchange	distance and area	microzone index	cells in footprint

Candidate location ranking - Istanbul analytical footprint



Analytical footprint, current estate and ranked candidate sites. The footprint is analytical, not an official district boundary.

06 / Data quality controls

46	44	2	0
checks executed	passes	non-critical warnings	failures / critical failures

Checks cover missing values, invalid and empty geometry, CRS, coordinate bounds, duplicate identifiers/points and spatial-join coverage. Critical failures block the run. Warnings require documented disposition.

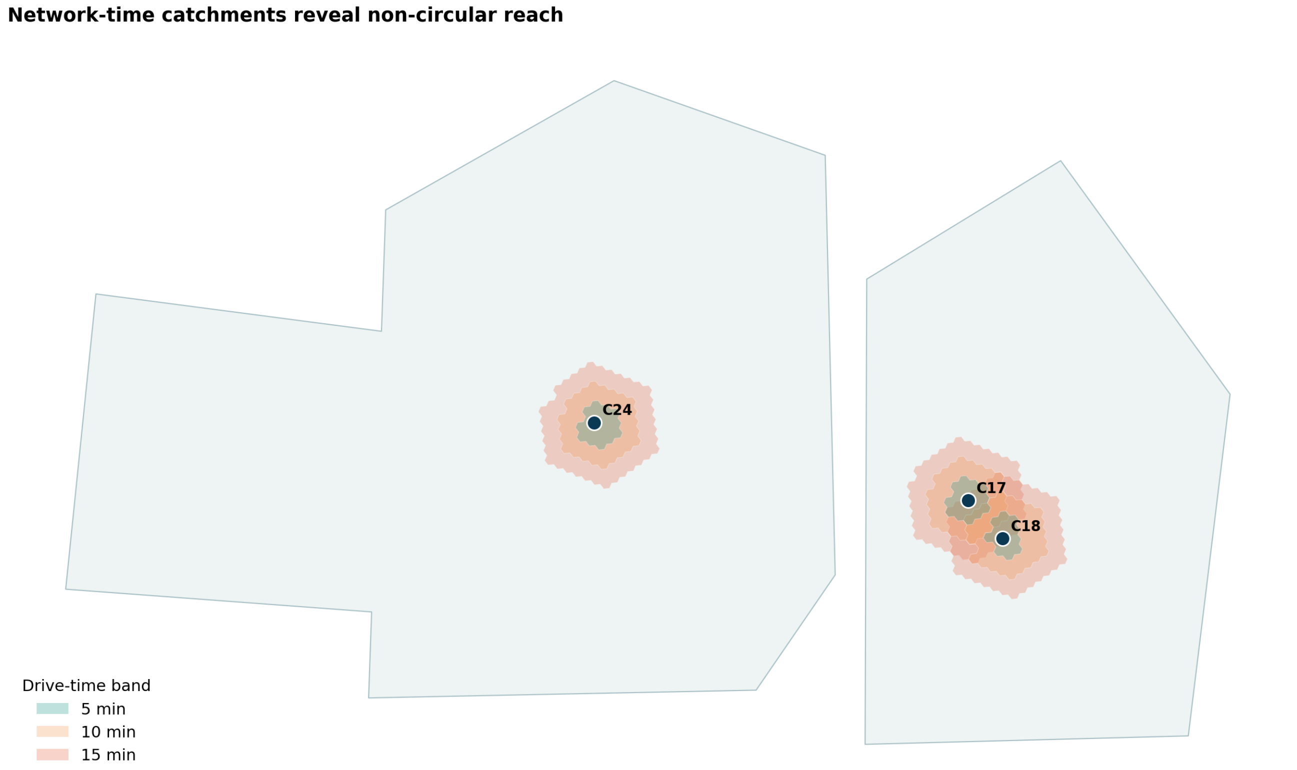
Open warnings

Dataset	Check	Status	Observed	Severity
competitors	spatial_join_unmatched	WARN	7	warning
pois	spatial_join_unmatched	WARN	16	warning

Seven synthetic competitors and sixteen synthetic POIs fall outside the analytical footprint. They are retained to exercise spatial-coverage warnings and do not enter in-footprint counts.

07 / Network accessibility and isochrones

A connected H3-adjacency graph has 5,965 nodes and 17,431 edges. Projected edge length is converted to time using mode-specific speeds and synthetic mobility indices. Dijkstra shortest paths produce 5/10/15-minute drive and walk catchments.



Network catchments for the leading candidate. Cells are reached by travel-time cost, not by a circular radius.

118,859	396,396	823,788
C24 five-minute drive population	C24 ten-minute drive population	C24 fifteen-minute drive population

08 / Straight-line versus network reach

A Euclidean buffer assumes uniform movement in every direction and can cross water, barriers and disconnected streets. Network time accumulates impedance along reachable links. The project stores both the network result and a naive comparison gap.

For C24 Ikitelli, the recorded naive-buffer versus 10-minute network population gap is 66.111%. This is why every dashboard labels network accessibility explicitly.

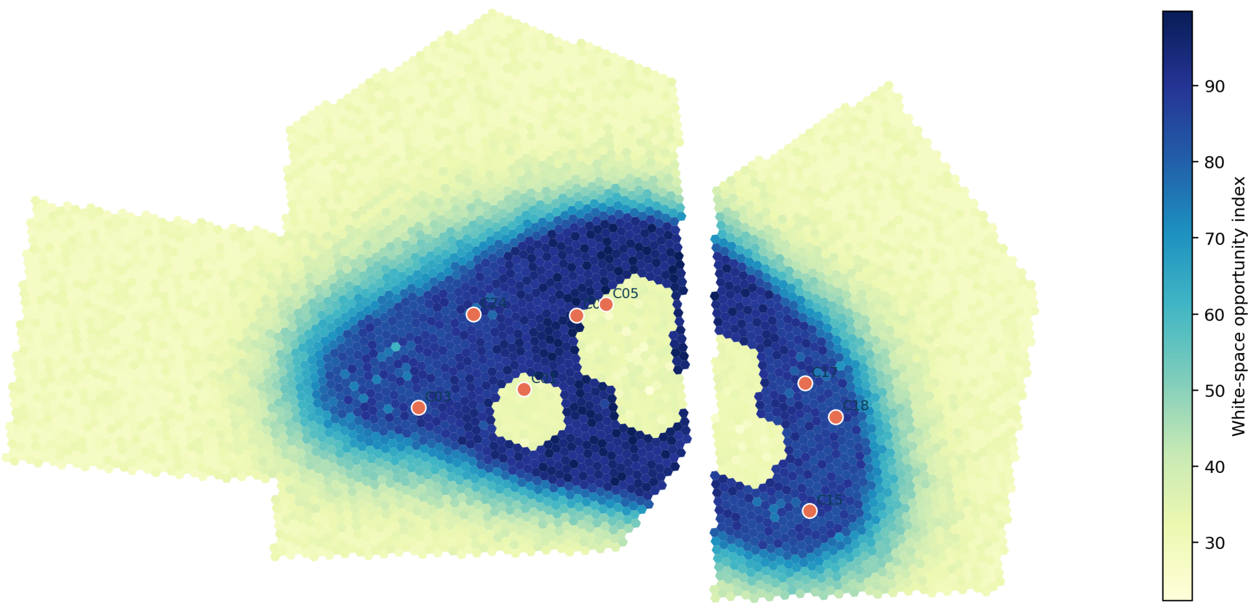
Method	Strength	Failure mode	Decision use
Circular buffer	Fast, simple benchmark	Ignores barriers and impedance	QA comparison only
H3 network time	Unique travel-time reach	Approximate speeds/links	Analytical screening
Live road routing	Turn/traffic aware	Source availability and variability	Pre-investment gate

Required independent validation

- rebuild drive and walk catchments with a current road network;
- test peak/off-peak traffic and directionality;
- verify bridge, ferry, toll and pedestrian restrictions;
- confirm the exact parcel access point rather than the candidate centroid.

09 / Competition, POIs and white space

White-space opportunity across H3 microzones



Microzone opportunity combines demand, competition headroom and current-estate coverage penalties.

Competitors are counted within a projected 3 km radius and POIs within 2 km. White space is not simply low competition: it rewards demand and uncovered reach while penalizing existing coverage and diversion.

> **A high white-space score is a screening signal. It does not prove parcel availability, commercial viability or absence of planned competition.**

10 / Huff gravity and cannibalization

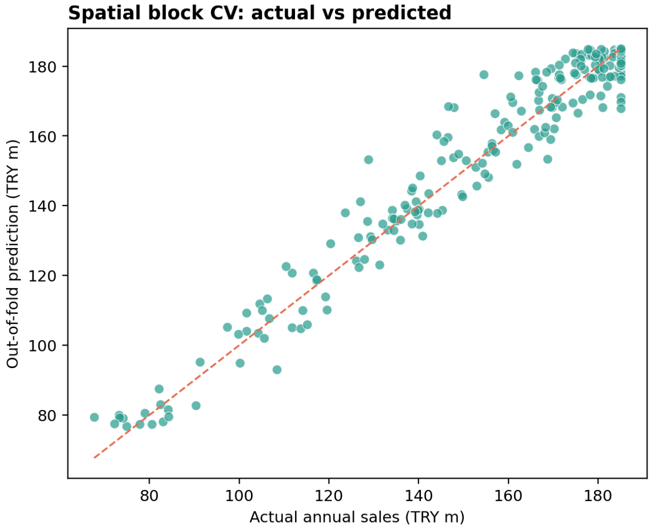
Huff share is proportional to candidate attractiveness and inverse distance with decay 1.65. Captured demand and diversion from the current estate are reported separately. Cannibalization combines 10-minute overlap and nearest-store distance.

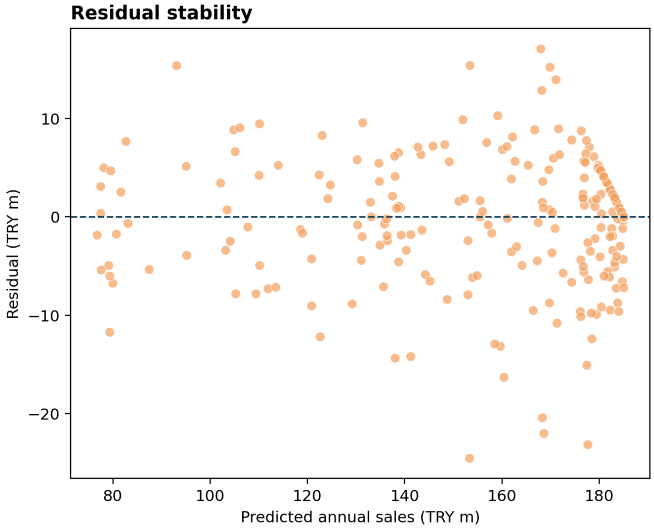
ID	Candidate	Huff diversion	Cannibalization	Nearest store km
C24	Ikitelli Industry	33.7%	1.0%	12.5
C18	Sultanbeyli Square	49.3%	1.8%	10.4
C17	Sancaktepe Hub	61.6%	2.6%	9.1
C07	Eyupsultan Hub	63.1%	34.9%	4.8
C03	Avcilar University	31.8%	1.0%	12.6
C05	Kagithane Axis	70.9%	61.0%	3.0
C01	Bahcelievler Metro	81.5%	53.5%	2.5
C15	Pendik Marina	51.0%	2.3%	9.5

C24 has 1.0% cannibalization and is 12.5 km from the nearest current store. C07 has 34.9% cannibalization at 4.8 km and therefore requires an estate-overlap review.

11 / Demand model and spatial validation

The 320-row benchmark is grouped into 33 projected 12 km blocks. Five GroupKFold splits ensure that neighboring sites in the same block cannot leak across training and validation. Final training occurs only after out-of-fold evaluation.





Out-of-fold actual-versus-predicted performance and residual stability.

3.819m	5.732m	0.964	2.65%
OOF MAE, TRY	OOF RMSE, TRY	OOF R2	OOF MAPE

High fit is expected for a deterministic synthetic target and must not be generalized to real store performance.

12 / SHAP demand explanations

C24 model baseline is TRY 162.787m and prediction is TRY 183.107m. Tree SHAP decomposes the difference into local feature contributions.

Feature	SHAP contribution, TRY m
accessible pop drive 10	+17.407
walkability index	+3.097
cannibalization risk	-1.158
competitor density 3km	+0.414
transit index	+0.253
poi density 2km	+0.141
income index	+0.138
rent try sqm month	+0.031

SHAP explains the demand model, not the final location score. AHP factor contributions provide the independent score-level audit.

SHAP values are associative and local to the fitted model. They do not establish causality and can reflect correlated synthetic features.

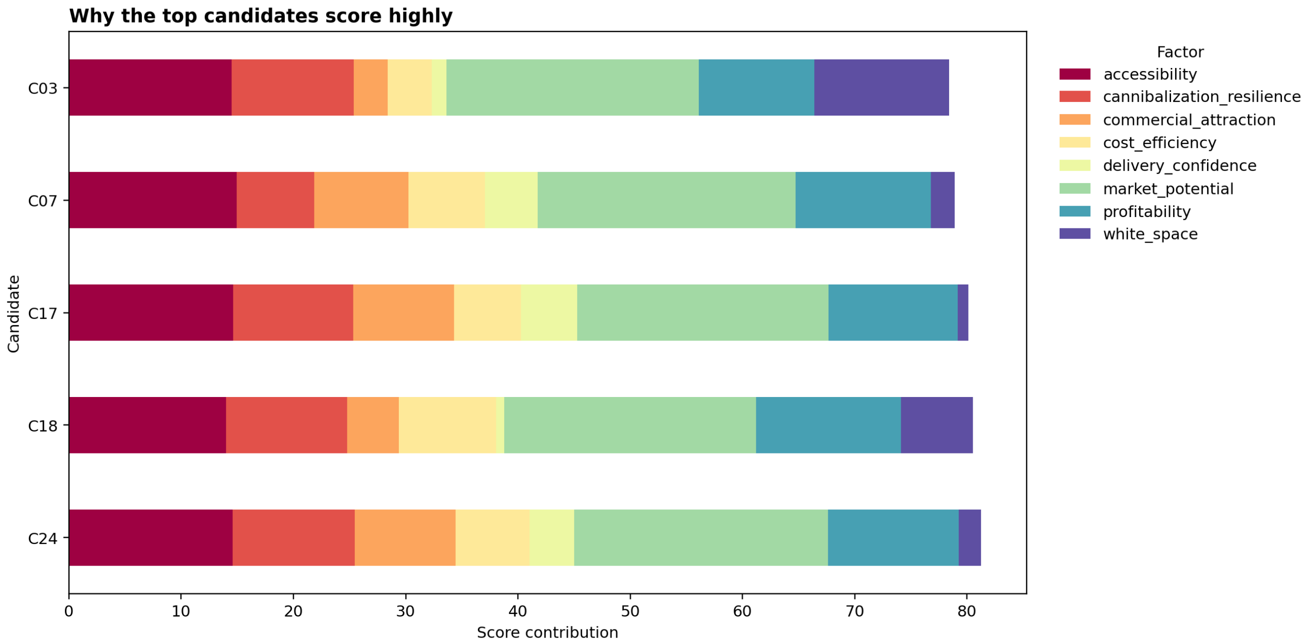
13 / AHP location scoring

AHP supplies transparent weights. Every factor has an explicit definition, direction, normalization and contribution. The reciprocal matrix has a consistency ratio of 5.399e-16, below the 0.10 threshold.

Factor	Direction	Normalization	Weight
market_potential	higher	min-max	23%
accessibility	higher	min-max	15%
commercial_attraction	higher	min-max	11%
white_space	higher	min-max	12%
cost_efficiency	higher	min-max	10%
cannibalization_resilience	higher	min-max	11%
profitability	higher	min-max	13%
delivery_confidence	higher	min-max	5%

Score = 100 x sum(weight x normalized factor). Raw values, normalization and candidate-level contributions are persisted in separate audit tables.

14 / Factor contributions and sensitivity



The top five scores are sums of visible factor contributions.

Candidate	Mean rank	P05	P95	Top-five probability	#1 probability
C24	1.67	1	3	100.0%	53.5%
C18	2.63	1	5	97.1%	30.9%
C17	3.13	2	5	95.7%	2.7%
C07	4.35	3	6	94.3%	0.0%
C03	4.86	1	9	55.5%	10.4%
C05	5.52	2	9	46.8%	2.5%

C24 remains top-five in 100% of 750 sampled weight sets. Stability means the ranking is not fragile to modest weight changes; it does not mean investment-ready.

15 / Location-allocation optimization

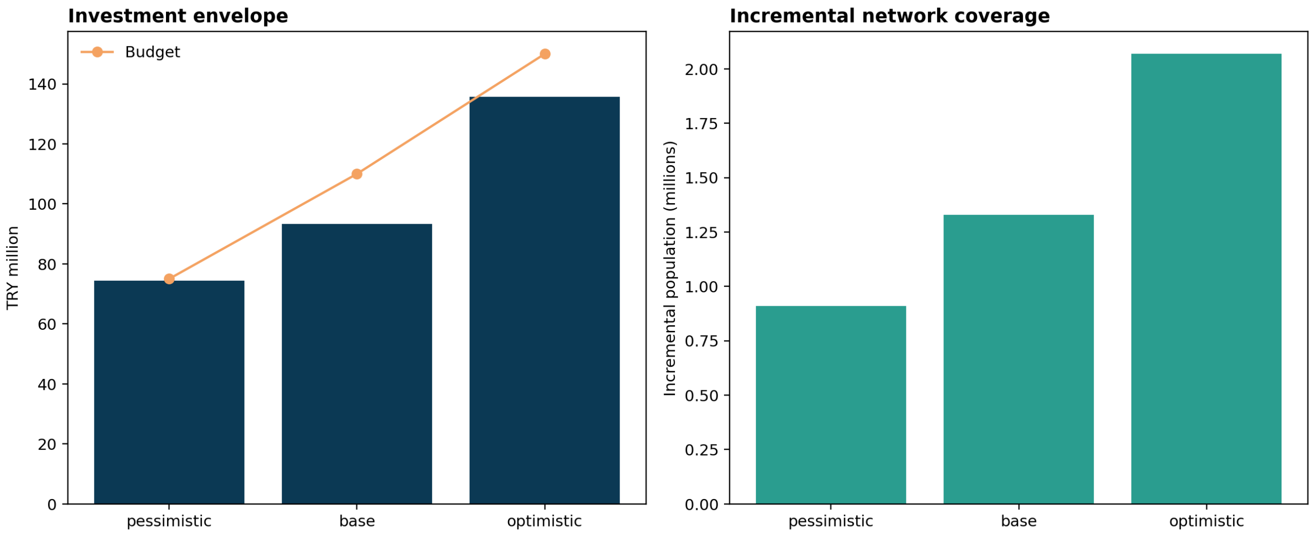
Binary candidate variables are combined with microzone coverage variables. The objective balances expected EBIT, location score and newly covered population, with a cannibalization penalty.

Element	Implementation
Budget	scenario-specific cap on opening cost
Minimum distance	pairwise conflict constraints
Capacity	maximum portfolio count by scenario
Coverage	10-minute drive microzone counted once
Feasibility	publish only terminal Optimal status
Objective	economics + score + incremental coverage - cannibalization

Why optimization follows ranking

- Overlapping high-score catchments can waste portfolio budget.
- A candidate may violate minimum distance from another candidate.
- A lower-ranked site can add more unique population coverage.
- Scenario economics can change the feasible site count.
- The optimizer does not waive field or investment constraints.

16 / Scenario results

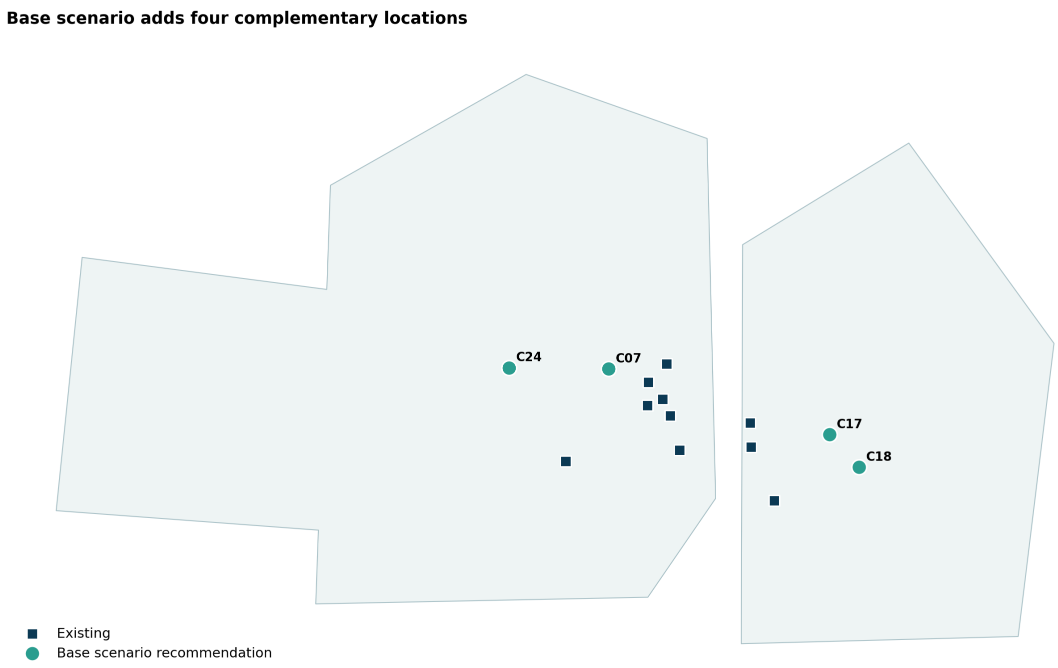


Scenario portfolios are comparative stress tests, not probability-weighted forecasts.

Scenario	Sites	Budget TRY m	Coverage	Sales TRY m	EBIT TRY m
optimistic	6	135.783	27.835%	1256.350	81.520
base	4	93.382	23.125%	732.554	28.826
pessimistic	3	74.473	20.447%	466.551	-2.071

The pessimistic portfolio remains feasible but its controlled expected EBIT turns negative (-TRY 2.071m), making downside diligence a required approval gate.

17 / Base portfolio and site economics



Existing stores and the four base-selected candidate corridors.

ID	Site	Score	Cost TRY m	Sales TRY m	3y ROI
C24	Ikitelli Industry	81.2	23.440	183.107	-16.0%
C18	Sultanbeyli Square	80.5	22.296	182.320	14.7%
C17	Sancaktepe Hub	80.1	24.261	182.183	-20.1%
C07	Eyupsultan Hub	78.9	23.385	184.944	-6.6%

Only C18 has positive controlled three-year ROI. Binding rent, fit-out, staffing, margin and opening-ramp evidence must replace synthetic assumptions before approval.

18 / PostGIS, API and observability

Capability	Implementation	Control
Spatial storage	PostGIS geometry and geography	WGS84 checks and GIST indexes
Analytics	SQL views for ranking, coverage and scenarios	source-to-view reconciliation
Service	FastAPI candidate/score/scenario endpoints	Pydantic bounds; read-only artifacts
Monitoring	Prometheus metrics and Grafana dashboard	health, latency, errors and drift
Deployment	Docker Compose and Kubernetes starter	read-only, cap drop, network policy

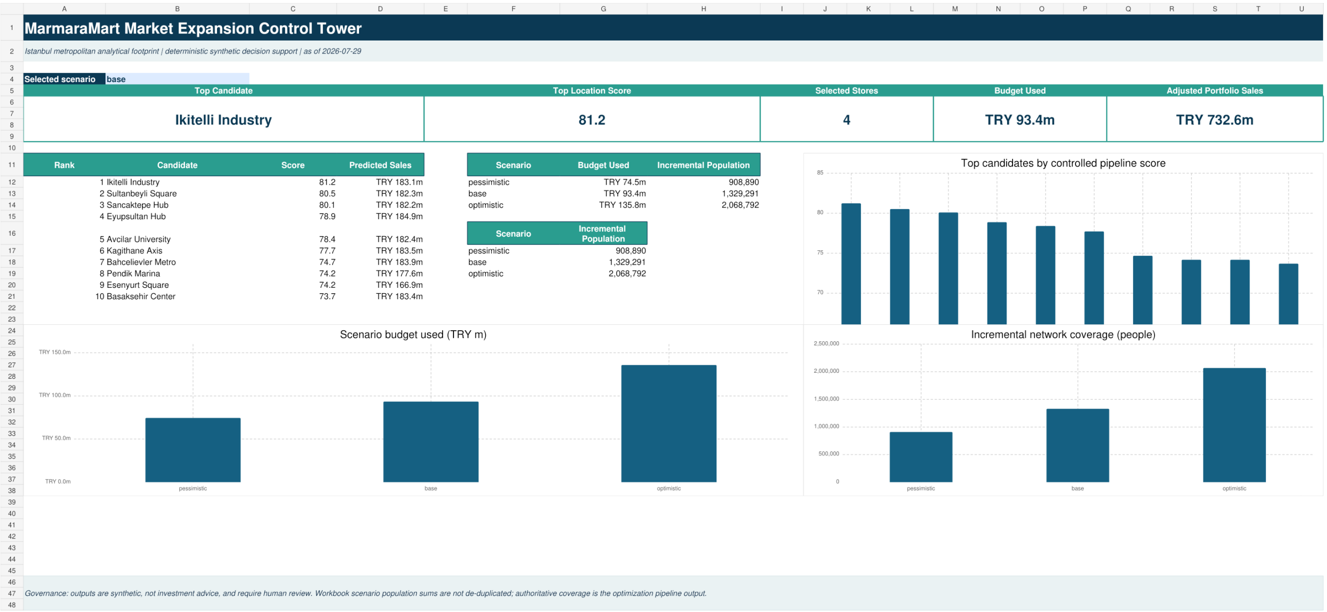
Service contract

- GET /health reports artifact readiness.
- GET /v1/candidates and /v1/candidates/{id} expose ranked evidence.
- POST /v1/score recomputes bounded, normalized weight scenarios.
- POST /v1/scenarios/evaluate returns controlled precomputed portfolios.
- GET /metrics exposes service indicators without sensitive payloads.

The included infrastructure is a production-oriented starter. No cloud deployment or live availability claim is made in Stage 1.

19 / Decision products and BI design

The same pipeline outputs feed interactive HTML maps, a formula-driven Excel scenario workbook, a PBIP/PBIR/TMDL starter, an executive deck and this report.



Dashboard design reference with ranking, scenario and factor-contribution views.

Deliverable	Validation completed	Remaining gate
Excel	formula scan and visual sheet inspection	business user acceptance
PowerPoint	all-slide render and overflow test	presenter review
PDF	all-page render and structural checks	reviewer sign-off
HTML maps	content, layers and attribution checks	browser/accessibility review
PBIP starter	JSON structure and source paths	Power BI Desktop render/refresh

20 / Governance, monitoring and incidents

Gate	Owner	Required evidence
Data	Data owner	source/license, contract, CRS and quality ledger
Model	Analytical owner	spatial CV, baseline, residuals and model card
Decision	Business + analytical	contributions, sensitivity and constraints
Field	Business/real estate	parcel, routing, legal and commercial diligence
Capital	Investment committee	approved business case and risk acceptance

Monitoring and incident priorities

- stop publication on any critical data-quality failure;
- track MAE/RMSE, feature distributions, SHAP and rank stability;
- surface non-Optimal optimizer status and block stale results;
- verify artifacts by SHA-256 before release and rollback;
- withdraw any report that may change candidate ranking or portfolio selection.

> **The final capital decision remains a human accountability and cannot be delegated to the scoring API, dashboard or optimizer.**

21 / Bias, threat model and open risks

Risk	Mechanism	Primary control
Affluence bias	income/purchasing-power proxy	sensitivity and service-equity overlay
Centrality bias	transit and POI advantage	white-space/cost balance and corridor review
Estate bias	current stores define coverage	report uncovered demand separately
Analytical tampering	weights/output edited after review	CODEOWNERS, manifest and protected main
Supply chain	dependency/image compromise	Dependabot, pip-audit, CodeQL and Trivy
Decision misuse	synthetic output presented as fact	persistent labels and human stage gates

The most important residual risk is not a software exploit but decision misuse. The system can be technically correct under its assumptions while commercially wrong. Governance labels and field evidence are therefore core controls.

22 / Limitations and required next evidence

- Replace synthetic sales, demand, rent, capex and opex with governed aggregates and binding quotes.
- Replace H3 routing with a current turn-aware road and pedestrian network.
- Validate peak/off-peak traffic, parcel entrance and delivery access.
- Add zoning, permit, frontage, parking, visibility and planned infrastructure.
- Add competitor quality, openings, closures and future pipeline.
- Backtest on temporally and spatially held-out real store openings.
- Calibrate uncertainty intervals and monitor residuals by geography/store format.
- Review geographic equity and proxy effects with business/legal stakeholders.
- Open and refresh the PBIP starter in Power BI Desktop.
- Observe CI, CodeQL and security workflows to terminal results after publication.

No candidate should advance to capital approval until its exact site, live routing and binding unit economics replace the synthetic assumptions.

23 / Source and license register

Source/component	Use	License / status
Project generator	commercial and performance data	original MIT; deterministic synthetic
Approximate anchors	geographic framing	manual approximate coordinates
OpenStreetMap tiles	interactive basemap	ODbL attribution included
H3	microzone indexing	Apache-2.0
GeoPandas / Shapely	spatial operations	BSD-3-Clause
scikit-learn / SHAP	model and explanations	BSD-3-Clause / MIT
PuLP / CBC	location allocation	MIT / EPL-2.0
Power BI Project format	PBIP/PBIR/TMDL starter	Microsoft product/docs terms

Primary references:
OpenStreetMap copyright and attribution
H3 documentation
GeoPandas project and license
Shapely documentation
Microsoft Power BI Projects overview

No external real demographic, transaction, mobility, rent or proprietary POI dataset is bundled in this package.

A / Appendix A - Candidate priority list

Rank	ID	Candidate	Score	Sales TRY m	Cost TRY m	Cannib.
1	C24	Ikitelli Industry	81.236	183.107	23.440	1.0%
2	C18	Sultanbeyli Square	80.521	182.320	22.296	1.8%
3	C17	Sancaktepe Hub	80.095	182.183	24.261	2.6%
4	C07	Eyupsultan Hub	78.880	184.944	23.385	34.9%
5	C03	Avcilar University	78.402	182.370	25.660	1.0%
6	C05	Kagithane Axis	77.712	183.456	21.967	61.0%
7	C01	Bahcelievler Metro	74.670	183.858	24.053	53.5%
8	C15	Pendik Marina	74.183	177.554	23.887	2.3%
9	C04	Esenyurt Square	74.170	166.920	23.259	0.3%
10	C06	Basaksehir Center	73.694	183.362	28.231	0.5%
11	C02	Beylikduzu Marina	72.787	169.162	23.812	0.2%
12	C20	Buyukcekmece Center	72.693	162.885	24.559	0.0%

The full 24-row ranking, all raw/normalized factors and economic fields are available in artifacts/data/candidate_scores.csv and the Excel workbook.

B / Appendix B - Reproducibility and QA evidence

Evidence	Location
Pipeline run summary	artifacts/metrics/pipeline_summary.json
Model metrics and folds	artifacts/metrics/model_metrics.json; artifacts/data/model_fold_metrics.csv
Out-of-fold predictions	artifacts/data/model_out_of_fold_predictions.csv
Data quality	artifacts/qa/data_quality_checks.csv
AHP specification	artifacts/data/factor_specification.csv
SHAP and score contributions	artifacts/data/shap_contributions.csv; score_contributions.csv
Scenario outputs	artifacts/data/scenario_summaries.csv; scenario_selections.csv
Tests and coverage	artifacts/qa/test_results.txt; coverage.json/xml
Artifact verification	artifacts/qa/artifact_verification.json
File hashes	MANIFEST.sha256; project_manifest.json

Pipeline run status: SUCCESS - seed 20260729 - runtime 9.570 seconds. Generated metrics are evidence, not manually authored presentation values.

The release package intentionally excludes Git metadata, caches and local inspection sidecars. GitHub publication is a separate, approval-gated stage.

C / Appendix C - Review decision record

Use this page to record the Stage 1 review outcome. Approval here means approval to publish the reviewed repository tree, not approval to invest in any location.

Review item	Reviewer / date / disposition
Analytical methodology	
Model validation and SHAP	
AHP weights and sensitivity	
Optimization constraints and scenarios	
Data quality and source/license	
Security, risk and operations	
Excel / maps / deck / PDF / PBIP	
GitHub publication authorization	

PROJECT_STATUS: READY_FOR_REVIEW - NOT_PUBLISHED

Publication requires the explicit repository link and authorization to publish on main. Until then, no remote, push, pull request or GitHub write is permitted.